

- 9 A cycle of changes in pressure, volume and temperature of gas inside a cylinder of a petrol engine is illustrated in Fig. 9.1. The gas is assumed to be ideal.

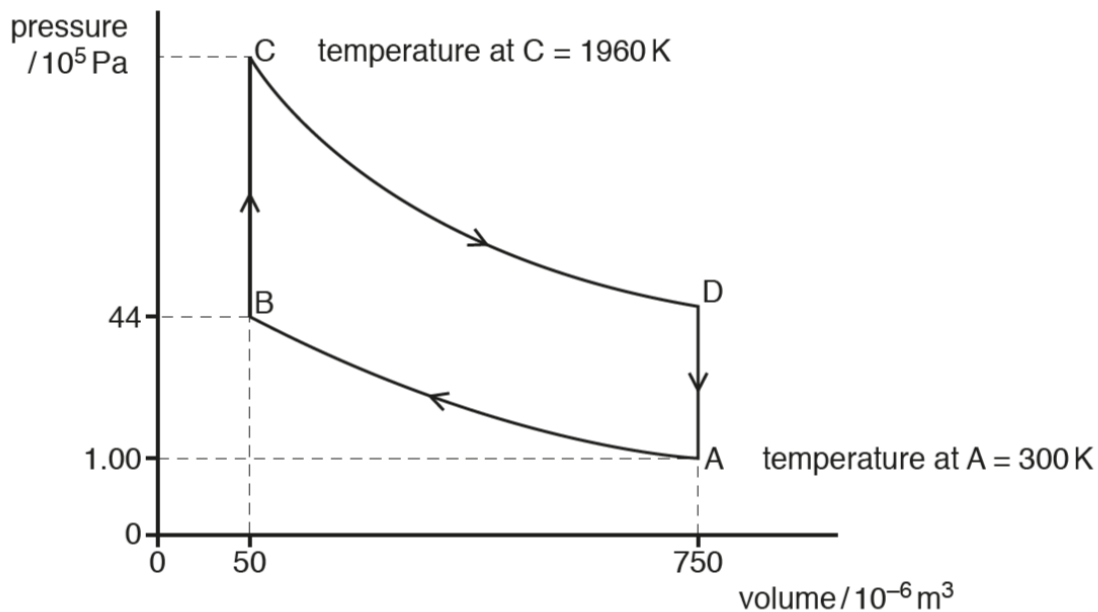


Fig. 9.1 (not to scale)

There are four stages in the cycle.

stage	description
A to B	Rapid compression of the gaseous petrol/air mixture with the temperature rising from 300 K at A. The pressure at B is 44×10^5 Pa.
B to C	The petrol/air mixture is exploded, resulting in an almost instant rise in pressure. At C the temperature is 1960 K.
C to D	Rapid expansion and cooling of the hot gases.
D to A	Return to the initial state of the cycle.

- (a) (i) Using appropriate values on Fig. 9.1, determine the number of moles present in the gases in the cycle.

$$\begin{aligned}
 pV &= nRT \\
 (1.00 \times 10^5)(750 \times 10^{-6}) &= n(8.31)(300) \\
 n &= 0.030 \text{ moles}
 \end{aligned}$$

number of moles = mol [2]

(ii) Calculate the temperature of the gas at B.

$$pV = nRT$$

$$(44 \times 10^5)(50 \times 10^{-6}) = n(8.31)(T_B)$$

$$T_B = 880 \text{ K}$$

or

$$\frac{p_A V_A}{T_A} = \frac{p_B V_B}{T_B}$$

$$\frac{(1.00 \times 10^5)(750 \times 10^{-6})}{300} = \frac{(44 \times 10^5)(50 \times 10^{-6})}{T_B}$$

$$T_B = 880 \text{ K}$$

temperature =K [2]

(iii) Calculate the pressure of the gas at C.

$$p_C V_C = nRT_C$$

$$(P_C)(50 \times 10^{-6}) = n(8.31)(1960)$$

$$p_C = 9.8 \times 10^6 \text{ Pa}$$

or

$$\frac{p_B}{T_B} = \frac{p_C}{T_C}$$

$$\frac{(44 \times 10^5)}{880} = \frac{p_C}{1960}$$

$$p_C = 9.8 \times 10^6 \text{ Pa}$$

pressure =Pa [2]

(iv) State

1. the numerical value of work done by the gas from B to C,

$W_{BC} = 0$ [B1] [1]

2. what is represented by the area ABCD enclosed by the graph.

It is the net work done [B1] BY the gas [B1] in one complete cycle
..... [2]

(b) Complete Table 9.1, which shows the work done on the gas, the heat supplied to the gas and the increase in internal energy of the gas, during the four stages in the cycle.

Table 9.1

stage	work done on gas /J	heat supplied to gas /J	increase in internal energy of gas /J
A to B	+ 360	0	360
B to C	0	+ 670	670
C to D	- 810	0	- 810
D to A	0	- 220	- 220

[4]

- (c) The efficiency of this engine is the ratio of the net work done by the gas to the heat supplied to the gas. Calculate the efficiency of this cycle.

$$\begin{aligned}\text{Efficiency} &= \frac{\text{work done by gas}}{\text{heat supplied}} \\ &= \frac{810 - 360}{670} \\ &= 67\%\end{aligned}$$

efficiency = [1]

- (d) Using the First Law of Thermodynamics, explain whether the r.m.s. speed of the molecules of the gas will increase, decrease or remains the same when the gas expands rapidly from C to D.

In stage C to D, negative work is done on the gas (gas is expanded) AND no heat is supplied or removed from the gas (expansion is rapid). The change in internal energy is negative, meaning the internal energy DECREASES. [first law concluded DECREASE in U: B1]

This is an ideal gas, so the internal energy is equal to the microscopic kinetic energy (microscopic potential energy is zero) so the kinetic energy of the molecules will decrease and thus the root mean square speed of the molecules will decrease. [B1]

- (e) Explain, in terms of the collision of the molecules of the gas with the walls of the container, why an expansion results in a change in the kinetic energy of the molecules from C to D.

In stage C to D the gas is expanded, walls of the container do NEGATIVE WORK [B1] on the gas molecules during collision.

This means energy is removed from the gas molecules, so the kinetic energy of the molecules decreases. [B1]

- (f) Calculate the total kinetic energy of the molecules of the gas at C.

$$\begin{aligned}\Sigma K &= N \left(\frac{3}{2} kT \right) \\ &= (0.030 \times 6.02 \times 10^{23}) \left(\frac{3}{2} \right) (1.38 \times 10^{-23}) (1960) \\ &= 732 \text{ J}\end{aligned}$$

total kinetic energy = J [2]

[Total: 20]

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